Configuring a QEMU based RISCV Virtual machine

Alan Johnson November 2024

The following steps were performed to configure a Fedora based VM running on a Linux system using Ubuntu[1] 24.10 with kernel 6.11. Different software revisions may have to be tweaked accordingly. The command below can be used to verify the Ubuntu software release and the Linux kernel version.

```
lsb_release -a
No LSB modules are available.
Distributor ID:     Ubuntu
Description:  Ubuntu 24.10
Release:      24.10
Codename:     oracular
$ uname -a
Linux beelink 6.11.0-9-generic #9-Ubuntu SMP PREEMPT_DYNAMIC Mon Oct 14 13:19:59 UTC
2024 x86_64 x86_64 x86_64 GNU/Linux.
```

Install the QEMU software

```
sudo apt install qemu-system-riscv64
sudo apt install u-boot-qemu opensbi
Note, selecting 'qemu-system-misc' instead of 'qemu-system-riscv64'
The following packages were automatically installed and are no longer required:
  linux-headers-6.11.0-8          linux-modules-extra-6.11.0-8-generic
  linux-headers-6.11.0-8-generic  linux-tools-6.11.0-8
  linux-modules-6.11.0-8-generic  linux-tools-6.11.0-8-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  qemu-system-misc

Setting up u-boot-qemu (2024.01+dfsg-5ubuntu2) ...
Setting up opensbi (1.5-1) ...
```

Create a directory called `riscv` in the home folder

```
mkdir ~/riscv
cd ~/riscv
```

Download the required files (Highlighted in Figure 2-1) from https://dl.fedoraproject.org/pub/alt/risc-v/disk_images/Fedora-Developer-38-20230519.n.0.SiFive.Unmatched.and.QEMU/

Or directly or with `wget`

```
wget https://dl.fedoraproject.org/pub/alt/risc-v/disk_images/Fedora-Developer-38-20230519.n.0.SiFive.Unmatched.and.QEMU/Fedora-Developer-38-20230519.n.0-u-boot.itb.xz
wget https://dl.fedoraproject.org/pub/alt/risc-v/disk_images/Fedora-Developer-38-20230519.n.0.SiFive.Unmatched.and.QEMU/Fedora-Developer-38-20230519.n.0-u-boot-spl.bin.xz
 wget https://dl.fedoraproject.org/pub/alt/risc-v/disk_images/Fedora-Developer-38-20230519.n.0.SiFive.Unmatched.and.QEMU/Fedora-Developer-38-20230519.n.0-sda.raw.xz
```

[1] Debian 12 was also tested successfully.

Install `xz` utilities

```
sudo apt install -y xz-utils
```

Unpack the Fedora `.xz` files in the riscv directory.

```
unxz Fedor*.xz
ls
Fedora-Developer-38-20230519.n.0-sda.raw     Fedora-Developer-38-20230519.n.0-u-boot-spl.bin Fedora-Developer-38-20230519.n.0-u-boot.itb
```

Create a script `startvm.sh` –

```
vi startvm.sh

qemu-system-riscv64 \
   -bios Fedora-Developer-38-20230519.n.0-u-boot-spl.bin \
   -nographic \
   -machine virt \
   -smp 2 \
   -m 8G \
   -device loader,file=Fedora-Developer-38-20230519.n.0-u-boot.itb,addr=0x80200000 \
   -object rng-random,filename=/dev/urandom,id=rng0 \
   -device virtio-rng-device,rng=rng0 \
   -device virtio-blk-device,drive=hd0 \
   -drive file=Fedora-Developer-38-20230519.n.0-sda.raw,format=raw,id=hd0 \
   -device virtio-net-device,netdev=usernet \
   -netdev user,id=usernet,hostfwd=tcp::3333-:22
```

In this case 8G of memory has been allocated, and port 3333 has been setup for ssh. The script should start the virtual machine as shown below:

```
./startvm.sh
```

```
U-Boot SPL 2023.04 (May 03 2023 - 00:00:00 +0000)
Trying to boot from RAM

U-Boot 2023.04 (May 03 2023 - 00:00:00 +0000)

CPU:   rv64imafdcsu
Model: riscv-virtio,qemu
DRAM:  8 GiB
Core:  35 devices, 15 uclasses, devicetree: board
Flash: 32 MiB
. . .
Kernel 6.2.16-300.0.riscv64.fc38.riscv64 on an riscv64 (ttyS0)

The root password is 'fedora_rocks!'.
root password logins are disabled in SSH starting Fedora 31.
User 'riscv' with password 'fedora_rocks!' in 'wheel' group is provided.

To install new packages use 'dnf install ...'

To upgrade disk image use 'dnf upgrade --best'

If DNS isn’t working, try editing ‘/etc/yum.repos.d/fedora-riscv.repo’.

For updates and latest information read:
https://fedoraproject.org/wiki/Architectures/RISC-V

Fedora/RISC-V
-------------
Koji:               http://fedora.riscv.rocks/koji/
SCM:                http://fedora.riscv.rocks:3000/
Distribution rep.:  http://fedora.riscv.rocks/repos-dist/
Koji internal rep.: http://fedora.riscv.rocks/repos/
fedora-riscv login:
```

Use `ssh` to login from the host machine username is `riscv` and password is `fedora_rocks!`

```
$ ssh -p 3333 riscv@localhost
The authenticity of host '[localhost]:3333 ([127.0.0.1]:3333)' can't be established.
ED25519 key fingerprint is SHA256:jkml56bGXAYeBrXgoj5orxHfX8bDvTVsSYlJxzvx/Mc.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:3333' (ED25519) to the list of known hosts.
riscv@localhost's password:
Last login: Mon Nov 11 15:40:47 2024 from 10.0.2.2
```

```
[riscv@fedora-riscv ~]$
```

Check the release

```
cat fedora-release
Fedora release 38 (Thirty Eight)
```

Check the kernel revision

```
uname -a
Linux fedora-riscv 6.2.16-300.0.riscv64.fc38.riscv64 #1 SMP Thu May 18 15:17:39 EDT
2023 riscv64 GNU/Linux
```

Check the virtual CPU

```
$ lscpu
Architecture:          riscv64
  Byte Order:          Little Endian
CPU(s):                2
  On-line CPU(s) list: 0,1
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0,1
```

Check the memory size

```
free -g
               total        used        free      shared  buff/cache   available
Mem:               7           0           4           0           3           6
Swap:              7           0           7
```

Resizing the image

Note it is possible to enlarge the virtual disk (if required) by the following steps:

Step 1

On the physical machine -

```
qemu-img resize Fedora-Developer-38-20230519.n.0-sda.raw +30G
WARNING: Image format was not specified for 'Fedora-Developer-38-20230519.n.0-sda.raw'
and probing guessed raw.
         Automatically detecting the format is dangerous for raw images, write
operations on block 0 will be restricted.
         Specify the 'raw' format explicitly to remove the restrictions.
Image resized.
```

Step 2

Resize the partition with `fdisk`[2].

```
fdisk Fedora-Developer-38-20230519.n.0-sda.raw

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): p
Disk Fedora-Developer-38-20230519.n.0-sda.raw: 38.98 GiB, 41850765312 bytes, 81739776
sectors
Units: sectors of 1 * 512 = 512 bytes
```

[2] Later versions of parted have removed this capability.

```
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x65f1d0f9

Device                                     Boot   Start      End  Sectors  Size Id Type
Fedora-Developer-38-20230519.n.0-sda.raw1 *       8192  1007615   999424  488M 83
Linux
Fedora-Developer-38-20230519.n.0-sda.raw2      1007616 17960959 16953344  8.1G 83
Linux

Command (m for help): e
Partition number (1,2, default 2): 2

New <size>{K,M,G,T,P} in bytes or <size>S in sectors (default 38.5G):

Partition 2 has been resized.

Command (m for help): p
Disk Fedora-Developer-38-20230519.n.0-sda.raw: 38.98 GiB, 41850765312 bytes, 81739776
sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x65f1d0f9

Device                                     Boot   Start      End  Sectors  Size Id Type
Fedora-Developer-38-20230519.n.0-sda.raw1 *       8192  1007615   999424  488M 83
Linux
Fedora-Developer-38-20230519.n.0-sda.raw2      1007616 81739775 80732160 38.5G 83
Linux

Command (m for help): w
The partition table has been altered.
```

Note An alternative to `fdisk` is `cfdisk`

```
sudo cfdisk Fedora-Developer-38-20230519.n.0-sda.raw
```

```
                    Disk: Fedora-Developer-38-20230519.n.0-sda.raw
                 Size: 28.98 GiB, 31113347072 bytes, 60768256 sectors
                          Label: dos, identifier: 0x65f1d0f9

   Device                                     Boot     Start      End  Sectors  Size Id Type
   Free space                                           2048     8191     6144    3M
   Fedora-Developer-38-20230519.n.0-sda.raw1  *         8192  1007615   999424  488M 83 Linux
>> Fedora-Developer-38-20230519.n.0-sda.raw2         1007616 60768255 59760640 28.5G 83 Linux

  Partition type: Linux (83)
 Filesystem UUID: 56f56257-a0d4-4865-a531-ade6042a110c
Filesystem LABEL: _/
      Filesystem: ext4

 [Bootable]  [ Delete ]  [ Resize ]  [  Quit  ]  [  Type  ]  [  Help  ]  [  Write ]  [  Dump  ]
```

Step 3

Power up the virtual machine and check the disk capacity.

```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           2.0G     0  2.0G   0% /dev/shm
tmpfs           779M  8.4M  770M   2% /run
/dev/vda2       7.9G  7.0G  782M  91% /
tmpfs           2.0G     0  2.0G   0% /tmp
/dev/vda1       441M  140M  287M  33% /boot
tmpfs           390M   16K  390M   1% /run/user/1000
```

Step 4

Grow the virtual disk partition device `/dev/vda2`

```
$ sudo resize2fs /dev/vda2
[sudo] password for riscv:
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/vda2 is mounted on /; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 4
The filesystem on /dev/vda2 is now 7470080 (4k) blocks long.
```

Step 5

Check the size again

```
[riscv@fedora-riscv ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           2.0G     0  2.0G   0% /dev/shm
tmpfs           779M  8.4M  770M   2% /run
/dev/vda2        28G  7.0G   21G  26% /
tmpfs           2.0G     0  2.0G   0% /tmp
/dev/vda1       441M  140M  287M  33% /boot
tmpfs           390M   16K  390M   1% /run/user/1000
```